

Detecting Crowded Trades Using Investor Embeddings from Institutional Holdings

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Introduction and Motivation

- **Key idea:** by treating portfolios like “documents,” investor and asset embeddings map investors and firms into a **latent space** where proximity reflects **economic similarity** (substitutes) and **strategy similarity**.
- **Application:** crowded trades amplify downside risk—when many investors follow similar strategies, exits can become **synchronized** and disorderly.
- **Measurement gap:** standard proxies (asset overlap, correlations, factor exposures) might be **imperfect** because they miss **latent strategy similarity**.
- **Usefulness:** a clustering-based crowding indicator can be used for **risk monitoring** and is directly testable via **coordinated selling** in filings and **next-quarter drawdowns/volatility**.

Asset Embeddings [Gabaix, 2025]

- The paper introduces “**asset embeddings**” and “**investor embeddings**”: vector representations of firms and investors learned from portfolio holdings.
- **Core theory claim:** under empirically realistic conditions, the holdings embed essentially all information relevant for prices.
- **Main empirical finding:** holdings-based embeddings substantially outperform standard firm characteristics on valuation and portfolio prediction benchmarks.

Research Question:

Can investor embeddings from holdings data identify time-varying clusters of similar strategies?

Possible sub-questions: Do embedding-based crowding indicators predict:

- ① *coordinated selling*
- ② *drawdowns/volatility*
- ③ *underperformance of crowded assets*

- **Holdings source:** WRDS FactSet Ownership Data.
- **Sample:** quarterly panel from 2005Q1 to 2022Q4 (quarterly frequency throughout the core holdings analysis).
- **Price/return data (for validation):** daily or monthly total returns for the same assets (plus market cap for weighting). This to test whether crowding predicts drawdowns, volatility, or underperformance of crowded positions.

Data: example

Example holdings snapshot (AQR Capital, 30 Sep 2025)

Company	Adj. MV
NVIDIA Corporation	4,158,237,186
Microsoft Corporation	3,231,184,977
Apple Inc.	2,755,127,665
Amazon.com, Inc.	1,537,753,884
Broadcom Inc.	1,416,338,600
Roblox Corp.	1,343,302,964
Meta Platforms Inc	1,272,505,758
Walmart Inc.	1,202,006,506
Alphabet Inc.	1,001,748,734
VeriSign, Inc.	985,063,497
Alphabet Inc.	954,655,356

Notes: Adj. MV in USD. Top holdings shown. Source: WRDS 13F Ownership Data.

- **Input: quarterly holdings panel.** Build an investor–asset matrix of (log) dollar holdings h_{ia} for each quarter; objective is to learn low-dimensional vectors for assets x_a and investors λ_i that explain cross-sectional holdings.
- **Three different approaches:**
 - ① **Recommender/PCA baseline:** Estimate $\{x_a, \lambda_i\}$ by minimizing squared errors with regularization: $h_{ia} \approx \lambda_i^\top x_a + \delta_i + \delta_a$
 - ② **Word2Vec:** Treat each portfolio as a “sentence”: order assets by portfolio weight, mask the k -th position, and train a shallow model to predict the masked asset from a context window.
 - ③ **OS-BERT and PS-BERT:** Convert portfolios into ordered token sequences and use masked-token prediction; learn contextualized representations via attention and add position embeddings to encode rank/order.
- **Outputs:** Each method produces embedding vectors.

- **Principle:** evaluate embeddings by **out-of-sample** performance on economically meaningful tasks.
- **Three benchmarks:**
 - ① **Relative valuation:** ridge regression of valuation residuals on embeddings.
 - ② **Return comovement:** use embeddings from $t-1$ to explain returns in t .
 - ③ **Managed portfolio prediction:** mask a large holding in test portfolios (e.g., $k=2$) and predict it.
- **Comparison discipline:** fixed train/test splits (firms or investors), ridge penalty via CV, and benchmarks vs **firm characteristics**.

- Make sure the embeddings are **L-2 normalized** in order to use cosine similarity to define proximity on the sphere.
- **Clustering models:**
 - ① Spherical k-means (Distance-based)
 - ② Mixture of von Mises–Fisher distributions (Probabilistic model)
- **Compute internal metrics on the sphere:**
 - ① Cosine-silhouette
 - ② Calinski–Harabasz index
 - ③ Davies–Bouldin index

- **Core idea:** crowding is not only “holding the same assets,” but investors following similar latent strategies.
- **Cluster concentration (counts):**

$$HHI_t = \sum_{c=1}^K s_{c,t}^2, \quad \text{where } s_{c,t} \text{ is the share of investors in cluster } c.$$

- **Tightness crowding:** average within-cluster cosine similarity.
- **Define thresholds:** flag a quarter as “crowded” if HHI_t is above the historical 90th (or 95th) percentile.
- **Track cluster dynamics:** compute transition matrices of investors across clusters over time.

- **Predictive (returns):** Does crowding at quarter t predict *worse next-quarter outcomes*?
 - Next-quarter **max drawdown** and **realized volatility** (from daily returns).
- **Unwind (holdings):** Do investors in the *crowded cluster(s)* reduce positions in the **same names** in the next filing?
 - Measure: synchronized decreases in portfolio weights / ownership among clustered investors.
- **Benchmark:** Compare to a simple **overlap-based** measure (e.g., common-ownership concentration).

Conclusion & Expected contributions

- **Thesis:** Investor embeddings from holdings can identify clusters of similar strategies and quantify **crowding**.
- **Method:** PS-BERT investor embeddings $\rightarrow$ spherical clustering (k-means / vMF) $\rightarrow$ crowding metrics (HHI + tightness).
- **Contributions:**
 - A **strategy-based crowding indicator** beyond simple holding overlap.
 - Evidence that crowding predicts **coordinated selling** and **worse next-quarter risk** (drawdowns/volatility).

Thank you

Thank you for your attention.

Questions?

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Gabaix, Koijen, R. Y. (2025).
Asset embeddings.
Working Paper.